

Density-guided two-stage small object detection in UAV images

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ABSTRACT

Object detection in Unmanned Aerial Vehicle (UAV) images is highly challenging due to the sparse distribution, local clustering of objects, and the small size of objects caused by long shooting distances, making it difficult to achieve accurate localization in high-resolution UAV images. Traditional methods struggle to capture small object features in dense regions, limiting their effectiveness in improving detection performance. To address these issues, we propose a Density-Guided Two-Stage Object Detection (DG-TSOD) framework for recognizing small objects. First, we design a dense object region seeker that focuses on high-density areas, enabling density-guided training and inference. Second, to enhance the accuracy of bounding box regression, we introduce the normalized Wasserstein distance as the loss function. This metric offers significant advantages for small object detection due to its scale invariance, smooth response to positional deviations, and ability to measure similarity between non-overlapping boxes. Finally, efficient multi-scale and deformable convolution modules are employed in the detection head to enhance multi-scale object localization without significantly increasing computational cost. Extensive experiments demonstrate that our method achieves state-of-the-art performance on the VisDrone, UAVDT, and TinyPerson datasets.

1. Introduction

As an emerging airborne platform, Unmanned Aerial Vehicles (UAVs) serve as a valuable complement to aircraft and satellites in remote sensing, owing to their flexibility, low cost, and multifunctionality. They play a significant role in applications such as intelligent traffic management (Han et al., 2024; Wang et al., 2025), disaster monitoring, crop growth assessment, and military reconnaissance. UAVs can significantly improve mission efficiency by acquiring ground scene images and performing object recognition and localization.

With the rapid advancement of Convolutional Neural Networks (CNNs) (Duan et al., 2019; Redmon et al., 2016; Ren et al., 2015) and Vision Transformers (ViTs) (Carion et al., 2020; Dosovitskiy et al., 2020; Liu et al., 2021b), object detection techniques have seen significant progress, leading to the emergence of numerous advanced detectors. These detectors achieve strong performance on natural image benchmarks such as MS-COCO (Lin et al., 2014) and PASCAL VOC (Everingham et al., 2010), but they encounter significant challenges when applied to UAV imagery. Unlike natural images, UAV imagery is captured from high altitudes and unique perspectives, resulting in a wide field of view, complex background, numerous small objects, and non-uniform distribution. As illustrated in Fig. 1(b), UAV datasets exhibit

significantly higher density and quantity of small objects. For instance, images in the MS-COCO and PASCAL VOC datasets typically contain 7 and 3 objects, respectively, while UAV datasets such as VisDrone (Du et al., 2019) and UAVDT (Du et al., 2018) contain an average of 53 and 18 objects. In addition, the average image widths in PASCAL VOC and MS-COCO are 500 and 640 pixels, respectively, while UAV images range from 1500 to 2000 pixels in width. Therefore, applying existing detection methods directly to high-resolution UAV images is challenging due to their higher object density and larger image size.

To narrow the performance gap between object detection in UAV and natural images, researchers have focused on capturing multiscale features (Lin et al., 2017a; Liu et al., 2021a; Singh & Davis, 2018; Yang et al., 2022), using contextual information (Bell et al., 2016; Chen & Gupta, 2017), and applying data augmentation techniques (Chen et al., 2020; Ghiasi et al., 2021; Jiang et al., 2021; Yu et al., 2020). However, progress remains limited due to the overwhelming number of small objects (less than 32×32 pixels) and the limited pixels they occupy, which provide insufficient visual information for robust feature representation. In addition, high-resolution images are difficult to process directly due to high computational cost and memory usage, and are typically rescaled to a standard resolution to meet detector input constraints. This rescaling, combined with feature downsampling in detectors, especially for

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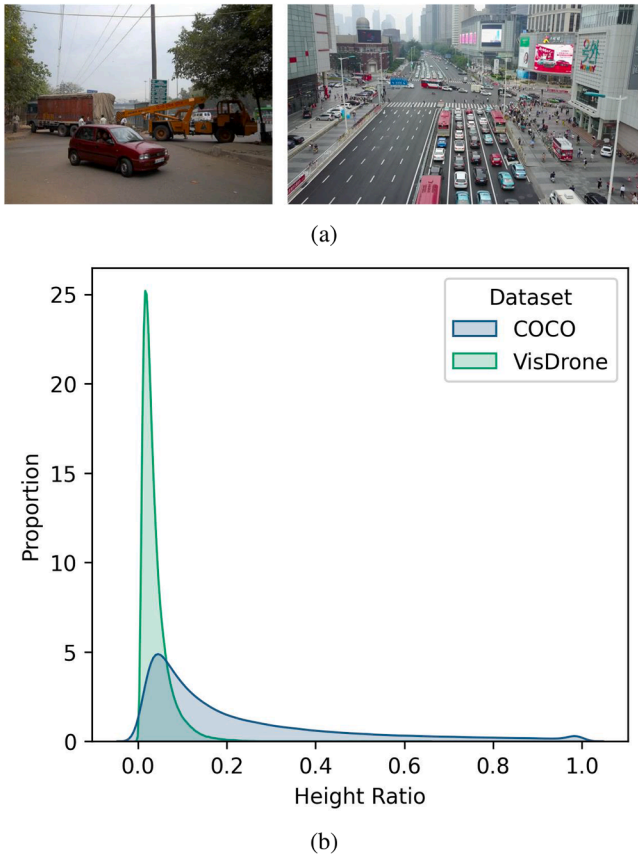


Fig. 1. (a) Comparison of instance scales between the COCO (left) and VisDrone (right) datasets. (b) Kernel density analysis of detection object height on VisDrone and COCO datasets.

small objects, often results in the loss of fine details and consequently reduces detection accuracy.

A common approach to address this problem is to divide the input image into uniform patches and upscale them (Ozge Unel et al., 2019). Although such an approach improves detection accuracy, it ignores the non-uniform distribution of objects and incurs substantial computational cost to process all patches. To overcome this limitation, recent methods have adopted object density-based cropping strategies. Density-based cropping dynamically adjusts the cropping region by prioritizing areas with higher object density. This strategy enhances cropping effectiveness and improves detection efficiency. Existing approaches often introduce additional learnable modules to extract dense object regions, such as density maps, clustering networks, or global-local detection frameworks (Meethal et al., 2023). Although these methods can focus on dense regions with small objects, they often rely on additional network components, increasing overall model complexity.

To improve small object detection accuracy while maintaining computational efficiency, we propose DG-TSOD, an efficient framework capable of adaptively identifying and localizing dense object regions. Specifically, an anchor-free detection network is adopted to regress object centers via the heatmap branch of the detection head and to locate dense regions based on the spatial distribution of these centers. Based on this, another branch equipped with efficient multi-scale and deformable convolution modules is employed to regress object width and height, thereby enhancing localization precision and size estimation accuracy. To further enhance performance, we employ the Normalized Wasserstein Distance (NWD) loss (Wang et al., 2021) as the regression loss, which effectively mitigates the performance degradation caused by localization deviations in small object detection. However, NWD is less effective for large objects compared to the $\mathcal{L}_1$ loss. To address this issue,

we combine the NWD and $\mathcal{L}_1$ losses using a weighting strategy that balances the advantages of both.

The main contributions of this paper are as follows:

(1) We propose a coarse-to-fine two-stage detection framework and a novel Dense Object Region Seeker (DOR Seeker), which identifies high-density regions by aggregating activated center points across channels on binarized heatmaps and performs grid-based clustering to guide refined detection.

(2) A weighted regression loss function combining Normalized Wasserstein Distance (NWD) and $\mathcal{L}_1$ loss is designed to improve robustness across different object scales, thereby boosting overall detection performance in UAV imagery.

(3) To address the severe occlusion of objects in dense regions of UAV images, we innovatively design an enhanced detection head that incorporates deformable convolution and an efficient multi-scale convolution module. This design enables adaptive perception of scale and shape variations, leading to a significant improvement in object localization accuracy.

Extensive experiments on three representative UAV image datasets demonstrate that the proposed method consistently outperforms existing approaches. Specifically, DG-TSOD achieves AP scores of 37.4 %, 21.4 %, and 61.5 % on the VisDrone, UAVDT, and TinyPerson datasets, respectively, surpassing current state-of-the-art methods.

2. Related works

This section aims to review representative methods and recent advances in the relevant research field, highlighting the strengths and limitations of existing works to clarify the motivation and innovations of our proposed approach.

2.1. Small object detection

Small object detection is a critical topic in computer vision and presents several technical challenges, including the loss of discriminative information, noisy feature representations, low robustness to bounding box perturbations, and long-tailed distributions of training samples relative to conventional object detection tasks. To address these issues, various strategies have been proposed. Multiscale feature learning is a common approach to handle scale variation. FPN (Lin et al., 2017a) integrates features from different backbone layers via a top-down pathway. SNIP (Singh & Davis, 2018) leverages image pyramids but retains only appropriate-scale samples. QueryDet (Yang et al., 2022) employs a cascaded sparse query mechanism to guide high-resolution feature computation based on the predicted coarse locations of small objects. By leveraging sparse convolutions, it reduces the computational burden of the detection head when processing low-level features. HRD-Net (Liu et al., 2021a) introduces a Multi-Depth Image Pyramid Network (MD-IPN) and a Multiscale Feature Pyramid Network (MS-FPN) to align and fuse multiscale features, mitigating the information imbalance across different scales and layers. In recent years, integrating contextual information has become an effective strategy to enhance small object detection performance. Context modeling is another effective direction. ION (Bell et al., 2016) combines internal and external ROI features. SMN (Chen & Gupta, 2017) is designed to capture instance-level context. It detects an object, stores it, and uses this prior knowledge to assist in detecting missed objects in subsequent iterations. In addition, data augmentation is widely used to enhance small object detection performance. Simple copy-paste (Ghiasi et al., 2021) is a powerful augmentation technique for instance segmentation and object detection, addressing various imbalance issues. Stitcher (Chen et al., 2020) uses the loss ratio of small objects to total loss as a feedback signal. When this ratio falls below a set threshold, it combines four images into one to increase the number of small objects in subsequent iterations. Yu et al. proposed SM (Yu et al., 2020) and SM+ (Jiang et al., 2021) as pre-training strategies to improve transfer learning by aligning object size

distributions between large-scale source datasets (e.g., MS COCO) and small-object datasets. Compared to the above methods, super-resolution techniques (Bai et al., 2018; Li et al., 2017) are more effective in enhancing small object details. However, most existing methods are designed for fixed viewpoints and general scene settings, whereas UAV imagery often presents more complex characteristics, such as drastic viewpoint variations, object occlusions, and sparse target distributions. In contrast to previous approaches, our work focuses on a density-guided mechanism that enables the detector to dynamically enhance detection accuracy in both dense and sparse regions, effectively addressing the limitations of existing methods in complex scenarios.

2.2. Object detection in UAV imagery

UAV imagery is typically captured at high resolutions. When using high-resolution images as input, it is common to extract multiple image patches from the original image for detection. Uniform cropping followed by zooming is a straightforward yet effective approach for detecting small objects. However, this approach often leads to many cropped patches containing only background, resulting in low detection efficiency and wasted computational resources.

To reduce unnecessary computation, ClusDet (Yang et al., 2019) introduces a clustering proposal network (CPNet) to locate potential object regions and discard background-only areas, thereby improving detection efficiency. In addition, a scale estimation network (ScaleNet) adaptively resizes the selected regions to match the detector's input size, reducing redundant computation. DMNet (Li et al., 2020) employs a lightweight CNN to generate a mask map and applies a sliding window and connected component analysis to produce clustering proposals. CDMNet (Duan et al., 2021) leverages a coarse-grained density map and a dual-task density estimation network to enhance small object detection, addressing distribution and scale adaptation issues. It significantly improves detection accuracy and computational efficiency through clustered region generation and advanced data augmentation strategies. UFPMP-Det (Huang et al., 2022) first uses a standalone Faster R-CNN to perform coarse-grained region detection, merges the resulting sub-regions into a unified image, and then applies RetinaNet for final detection. CZ Detector (Meethal et al., 2023) introduces a cascading zoom mechanism, where dense regions are localized and assigned new class labels during training. During inference, these dense regions are detected alongside base-class objects and subsequently passed to a second-stage detector. Furthermore, ESOD (Liu et al., 2024c) avoids redundant feature extraction by performing preliminary object localization and adaptive image slicing at the feature level using the proposed ObjSeeker and AdaSlicer modules. Therefore, such methods are often more complex to use in practice compared to the uniform cropping strategy. In this work, we also adopt a density-guided cropping strategy, but repurpose the detector itself to automatically locate dense regions, eliminating the need for additional learnable components or density estimation networks. As a result, our method maintains the simplicity and practicality of the uniform cropping approach while effectively preserving the performance benefits brought by density-guided detection.

3. Method

This section presents the proposed DG-TSOD framework, designed to address the unique challenges of object detection in UAV images. As illustrated in Fig. 2, during training phase, the ground-truth bounding boxes of each image are input into the proposed Dense Object Region Seeker (DOR Seeker) to extract dense object regions as a preprocessing step. Subsequently, we perform data augmentation by incorporating high-resolution versions of the extracted dense regions, along with their corresponding labeled bounding boxes, into the training set. This augmentation ensures that the training data includes detailed high-resolution representations of densely distributed small objects, enabling

the detector to better capture fine-grained object features. During inference, the detection process is divided into two stages. First, the image undergoes a coarse detection process to identify dense regions, which are subsequently cropped. Second, these crops are then processed by the same detector to perform fine-grained object detection. Finally, the coarse detection results from the original image are combined with the refined results obtained from the crops. In the following subsections, we detail the principles of the proposed DG-TSOD framework and the DOR Seeker.

3.1. DG-TSOD

The architecture of the proposed detector, DG-TSOD, is illustrated in Fig. 3. We adopt an anchor-free detection framework and choose HRNet (Wang et al., 2020) as the backbone network, primarily due to its outstanding performance in preserving high-resolution representations. Unlike traditional convolutional neural networks that progressively reduce feature map resolution through downsampling, HRNet maintains high-resolution feature maps via a multi-scale parallel architecture. This design enables the network to retain fine-grained image details, which is particularly beneficial for small object detection, as it helps preserve critical information that might otherwise be lost in low-resolution features. Moreover, HRNet's multi-resolution feature fusion mechanism provides rich contextual information, enhancing the model's ability to distinguish subtle differences between objects and background, further improving detection accuracy in complex scenes.

For an input image $I \in \mathbb{R}^{H \times W \times 3}$, it is first processed by the backbone and neck networks, resulting in a feature map $F \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$, where H and W denote the height and width of the input image, and C is the number of output feature channels. In the heatmap branch, to better localize the center points of small objects and prevent loss of critical spatial information due to downsampling, we apply Pixel Shuffle (Shi et al., 2016) operation to the output feature map. This operation upsamples the feature map by a factor of 2, effectively restoring it to the original image resolution. As a result, the model can perform detection on high-resolution features, thereby improving localization accuracy and overall detection efficiency.

Pixel Shuffle is an efficient upsampling technique widely used in image super-resolution and feature reconstruction tasks. Specifically, Pixel Shuffle rearranges the pixels in each channel of a low-resolution feature map into a higher-resolution spatial layout, thereby avoiding the information loss and computational overhead commonly associated with traditional interpolation-based upsampling methods. Given a low-resolution feature map of size $H \times W \times C_{in}$, the Pixel Shuffle operation increases its spatial resolution by a factor of r , resulting in an output of size $rH \times rW$. Simultaneously, the number of channels is reduced from C_{in} to:

$$C_{out} = \frac{C_{in}}{r^2}, \quad (1)$$

The upsampling process rearranges the pixel values in a low-resolution feature map to obtain a higher-resolution representation. The operation can be formulated as:

$$\begin{aligned} \text{PixelShuffle}(x) &= \text{reshape}(x, (H, W, r, r, C_{in})) \\ &\rightarrow \text{permute}(x, (H, W, C_{out})) \end{aligned} \quad (2)$$

where the reshape operation reshapes the input feature map according to the upsampling factor r , yielding a new spatial configuration. To enhance both computational efficiency and category-specific feature extraction, we adopt a grouped convolutional layer with the number of groups set to the number of object categories, i.e., $\text{group} = C$. Unlike standard convolution, where each output channel interacts with all input channels, grouped convolution decomposes the input feature map $X \in \mathbb{R}^{H \times W \times C}$ into C disjoint channel groups $\{X_1, X_2, \dots, X_C\}$, and performs independent convolutional operations within each group. For-

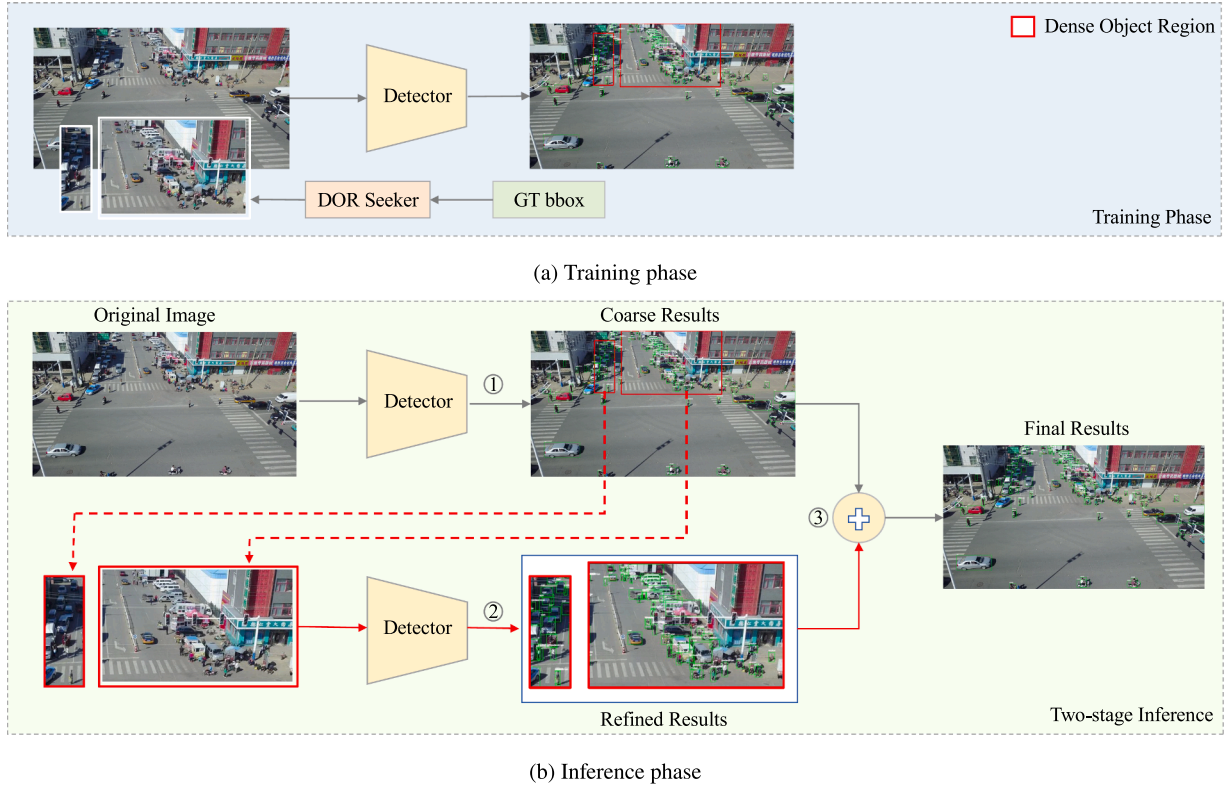


Fig. 2. Coarse-to-Fine object detection network. (a) Training phase: Ground-truth boxes are used by the DOR Seeker to extract dense object regions. These regions are augmented as high-resolution crops to enhance the learning capability of the detector. (b) Inference phase: Coarse detection is performed on the original image, while dense regions are simultaneously located. These regions are cropped and reprocessed by the same detector for refined detection. Final results are obtained by fusing both coarse and refined outputs.

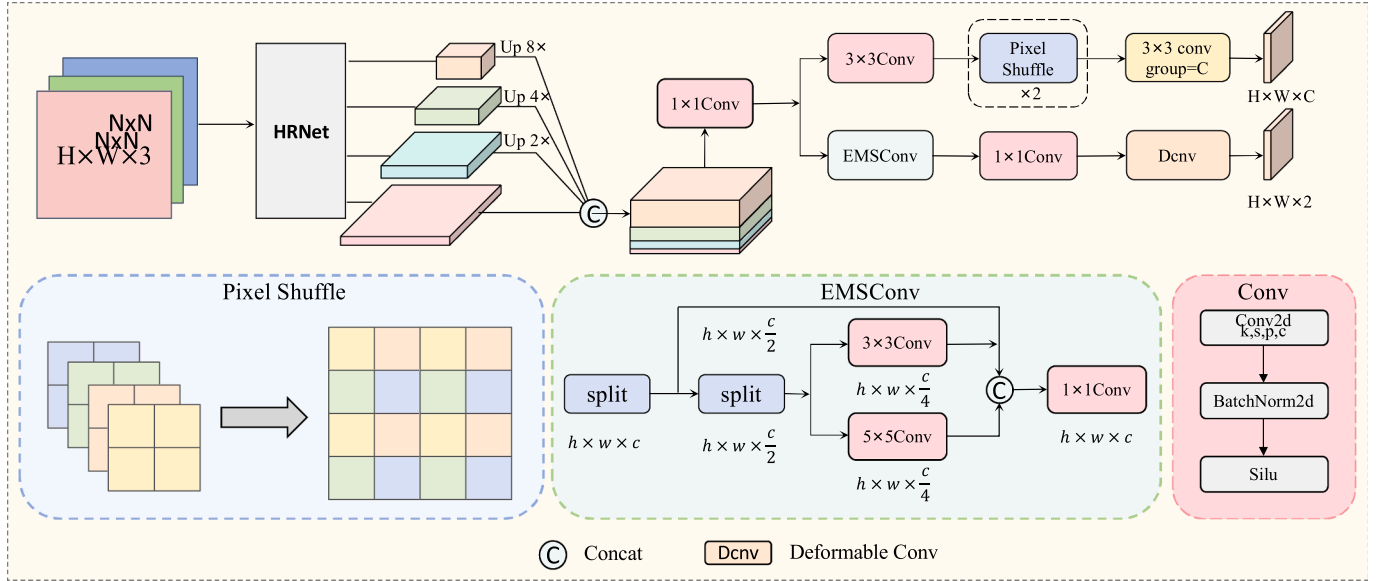


Fig. 3. Overview of DG-TSOD , consisting of three specialized modules: Pixel Shuffle, Efficient Multi-Scale Convolution (EMSConv), Deformable Convolution(Dcnv).

maximally, the output for the i th group can be expressed as:

$$\mathbf{Y}_i = \mathbf{W}_i * \mathbf{X}_i, \quad i = 1, \dots, C, \quad (3)$$

where $*$ denotes the convolution operation and $\mathbf{W}_i$ is the filter set for the i th group. This design allows each group to specialize in modeling features associated with a specific category, thereby reducing inter-class interference and improving the discriminability of learned

representations. Meanwhile, by limiting the connectivity between input and output channels, the grouped convolution reduces the number of parameters and floating-point operations by a factor of $1/C$, achieving a favorable balance between expressiveness and efficiency.

In the regression branch, we introduce two modules: the Efficient Multi-Scale Convolution (EMSConv) module (Rahman et al., 2024) and the Deformable Convolution(Dcnv) module (Zhu et al., 2019), which

together enhance the network's ability to accurately localize objects. The EMSConv module first applies a split operation to divide the input feature map along the channel dimension into two parts. One part is directly used to preserve fine-grained features for final feature fusion. The other part is further split along the channel axis, with each split processed independently using convolutional kernels of different sizes (e.g., 3×3 and 5×5) to capture multi-scale information. Finally, the outputs of these parallel convolutions are concatenated along the channel dimension and passed through a 1×1 convolution to fuse the features and generate the final output.

3.2. DOR Seeker

To enhance the detector's performance in small object detection, it is crucial to extract high-quality dense object regions. We define the quality of such regions based on the following constraints: (1) they should contain as many small objects as possible; (2) they should be easily locatable during inference; (3) their total number should be optimal to minimize computational overhead. Recent studies have attempted to introduce additional networks to select dense regions from the input image, followed by conventional object detection. However, the increased computational burden introduced by these auxiliary networks often limits overall performance. In contrast, we propose a Dense Object Region Seeker (DOR Seeker) module that directly utilizes the center heatmap generated by the heatmap branch to identify dense regions. This module adaptively searches for dense regions based on high-resolution heatmaps and adjusts their sizes to fit the detector's input. The entire process is unsupervised and lightweight, making it significantly more efficient and easier to implement.

First, the center heatmap generated by the heatmap branch is binarized using local maximum suppression, resulting in a binary heatmap $B \in [0, 1]^{H \times W \times C'}$, where C' denotes the number of object categories. The binarized heatmap is then divided into a 16×10 grid of non-overlapping equal-sized cells, indexed as (i, j) , where $i \in [1, 16]$ and $j \in [1, 10]$. For each grid cell (i, j) , the total number of center points across all C' channels is counted, and a fixed threshold is set. The 16×10 grid division corresponds to a resolution of 1024×640 pixels, with each cell covering a 64×64 pixel region. This design is motivated by the COCO benchmark standard, where objects within 32×32 pixels are categorized as small-scale. By setting the grid size to 64×64 , we ensure sufficient spatial granularity for dense target detection. Empirical analysis shows that when a grid cell contains two or more center points (i.e., density threshold $n \geq 2$), the probability of target overlap increases significantly, effectively identifying high-density regions.

If the total count in a cell exceeds the threshold, the corresponding cell index is retained. Then, a connected component clustering algorithm is applied to merge spatially adjacent candidate cells into larger candidate regions. All candidate regions are sorted based on their area in descending order, and the top K regions with the largest objects are selected. To prevent large targets in the image from being inadvertently cropped, each selected region is further enlarged by a factor of 1.2 before cropping. This results in the final set of K dense image crops used for refined detection.

The detailed procedure of the proposed DOR Seeker is illustrated in Algorithm 1.

3.3. Two-stage inference

The first stage of the inference process follows a standard object detection framework to generate coarse detection results. Simultaneously, the DOR Seeker identifies the top- K dense object regions based on the center heatmap produced by the detector's heatmap branch. In the second stage, the same detector architecture is reused; however, its input consists of the enlarged versions of the dense regions obtained in the first stage. During this stage, the standard inference procedure is reapplied with the goal of performing refined detection within the dense re-

Algorithm 1 Dense Object Region Seeker.

Input: B : Binarized heatmap.

Output: D : list of Density crop coordinates.

Parameters: τ : the threshold of central points within a grid

K : the number of crops.

$w = \frac{W}{16}, h = \frac{H}{10}$

$D \leftarrow \{\}$

$G[i, j] = 0, \quad \forall i \in \{1, \dots, 16\}, \quad \forall j \in \{1, \dots, 10\}$

for $c = 1$ to 10 **do**

for $i = 1$ to 16 **do**

for $j = 1$ to 10 **do**

$G[i, j] = \text{sum}(B[w \times (i - 1) : w \times i, h \times (j - 1) : h \times j])$

end for

end for

end for

$\text{Grid} \leftarrow G[i, j] > \tau$

$D \leftarrow \text{eight-connected}(\text{Grid})$

$D \leftarrow \text{sort-by-area}(D)$

if $\text{len}(D) > K$ **then**

$D \leftarrow D[1 \dots K]$

end if

$D \leftarrow \text{scale}(D, 1.2)$

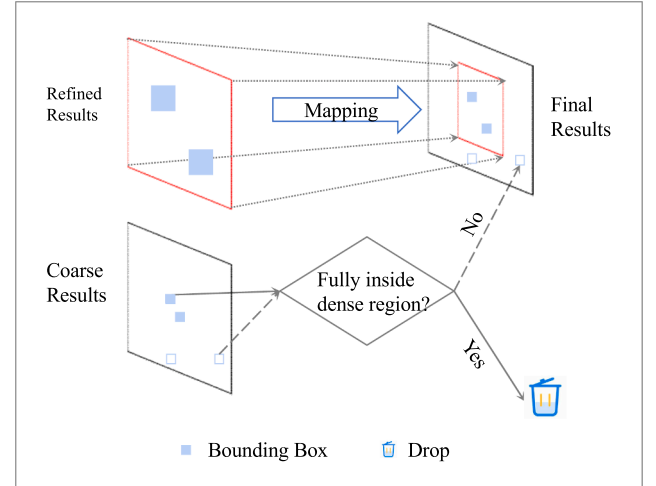


Fig. 4. Overview of the fusion mechanism.

gions. The core objective of this stage is to improve detection accuracy for objects within these crops, ensuring that objects are more accurately recognized under higher-resolution input.

Finally, to fuse the detection results from the first and second stages, the bounding boxes from the second stage must be mapped back from the crops coordinates to the coordinate system of the original image. As illustrated in Fig. 4, let the downsampling factor be denoted as S , and the coordinates of the crops in the original image be $(D_{x1}, D_{y1}, D_{x2}, D_{y2})$. The bounding boxes $\{d_i\}$, produced by the detector on the enlarged dense region are denoted as $(x_{1,i}, y_{1,i}, x_{2,i}, y_{2,i})$. To project these boxes back to the original image coordinate space, we obtain the final set of detection boxes $\{b_i\}$, by applying coordinate transformation and offset correction. Specifically, b_i is computed using Eq. (4):

$$b_i = (Sx_{1,i}, Sy_{1,i}, Sx_{2,i}, Sy_{2,i}) + (D_{x1}, D_{y1}, D_{x2}, D_{y2}) \quad (4)$$

During the results fusion stage, all coarse detection results from the original image are iterated over, and each bounding box is checked to determine whether it is fully contained within one or more crops. Specifically, a bounding box is considered fully contained if all of its boundaries (top, bottom, left, and right) lie entirely within the boundaries of a crop. If this condition is met, the corresponding detection box

is discarded and excluded from the final results. Only those coarse detection boxes not covered by any crops are retained and added to the final detection output, ensuring that redundant detections are avoided and results from the fine detection stage are prioritized.

3.4. Loss function

In our work, we employ two different loss functions to optimize the heatmap and regression branches. Inspired by CenterNet (Duan et al., 2019), for the heatmap branch, we adopt a modified version of Focal Loss (Lin et al., 2017b). Specifically, the heatmap branch generates a center heatmap $\hat{Y} \in [0, 1]^{H \times W \times C'}$, where each pixel value represents the likelihood of being the center of an object. When $\hat{Y}_{xyc} = 1$, the pixel at location (x, y) is exactly the center of an object. The closer it is to the object center, the higher the predicted value; conversely, $\hat{Y}_{xyc} = 0$ indicates background regions.

$$\mathcal{L}_h = -\frac{1}{N} \sum_{xyc} \begin{cases} (1 - \hat{Y}_{xyc})^\alpha \log(\hat{Y}_{xyc}) & \text{if } Y_{xyc} = 1 \\ (1 - Y_{xyc})^\beta (\hat{Y}_{xyc})^\alpha \log(1 - \hat{Y}_{xyc}) & \text{otherwise} \end{cases} \quad (5)$$

where α and β are the hyperparameters of the Focal Loss, set to 2 and 4 respectively, which help balance hard and easy examples as well as positive and negative samples. N denotes the total number of object center points in the image.

In object detection tasks, common approaches for regressing object width and height include directly predicting the center coordinates and dimensions of the bounding box, or optimizing the network by evaluating the overlap between predicted and ground truth boxes. These methods typically rely on IoU (Intersection over Union)-based loss functions, which guide the model by maximizing the overlapping area between predicted and ground truth boxes, with the goal of making the predictions as close as possible to the ground truth. However, IoU-based loss functions have several inherent limitations. First, they are only effective when the predicted and ground truth boxes overlap. Second, IoU losses are not sufficiently sensitive to changes in the scale of the bounding box, making them less effective in handling irregular or small objects.

To alleviate the sensitivity of IoU-based losses to positional deviations in small object detection, we adopt the Normalized Wasserstein Distance (NWD) as our regression loss. This method models bounding boxes as 2D Gaussian distributions and measures their similarity based on distributional distance. The final normalized formulation is defined as:

$$\text{NWD}(N_p, N_g) = \exp \left(-\sqrt{\frac{W^2(N_p, N_g)}{C}} \right), \quad (6)$$

where N_p and N_g denote the predicted and ground truth bounding box distributions, respectively. $W^2(\cdot)$ represents the squared Wasserstein distance between two distributions, C is set as a constant related to the dataset and empirically defined as the average absolute size of objects in the dataset. The final NWD-based loss function is formulated as:

$$\mathcal{L}_{\text{nwd}} = 1 - \text{NWD}(N_p, N_g). \quad (7)$$

We observe that in the early stage of model training, due to unstable optimization, the distributions of predicted and ground truth boxes may differ significantly. As the magnitude of the Wasserstein distance is proportional to this divergence, the value of W^2 tends to be large, resulting in a near-zero NWD value. Consequently, the model receives insufficient gradient feedback, particularly for large objects. To mitigate this issue, we combine the NWD loss with the $\mathcal{L}_1$ loss to enhance training stability and improve bounding box regression. The $\mathcal{L}_1$ loss is defined as:

$$\mathcal{L}_1 = \frac{1}{N} \sum_{i=1}^N |S_i - \hat{S}_i|, \quad (8)$$

where $S_i = (h_i, w_i)$ denotes the predicted height and width from the regression branch, and $\hat{S}_i \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 2}$ is the corresponding ground truth map.

The $\mathcal{L}_1$ loss exhibits linear gradients, where larger errors lead to stronger gradient signals. This property provides heavier penalties for large deviations, which helps the model adjust distributional differences more effectively during the early training stages and accelerates convergence. Therefore, we incorporate the $\mathcal{L}_1$ loss to enhance gradient strength. The final detection loss is defined as:

$$\mathcal{L} = \mathcal{L}_h + \lambda_1 \mathcal{L}_{\text{nwd}} + \lambda_2 \mathcal{L}_1, \quad (9)$$

where λ_1 and λ_2 are hyperparameters, which we empirically set to 2 and 0.5, respectively.

4. Experiments

To thoroughly assess the proposed approach, we conduct both quantitative and qualitative experiments across multiple datasets. These are followed by ablation studies that examine how different components influence the overall performance.

4.1. Datasets and protocols

We evaluate our proposed method on three public datasets: VisDrone, UAVDT, and TinyPerson (Yu et al., 2020). Below, we briefly introduce these datasets and their evaluation protocols.

VisDrone: This dataset consists of 8,599 images captured by UAVs, with 6,471 images used for training, 548 for validation, and 1,580 for testing. The image resolution is approximately 2000×1500 pixels. It includes 10 object categories, most of which represent vehicles and pedestrians observed from a UAV perspective. On average, each image contains 53 objects. Since the test set and evaluation server are not publicly accessible, we follow a common practice in existing literature by using the validation set to assess the performance of our method (Liu et al., 2024a).

UAVDT: This dataset comprises 50 video sequences, with 30 sequences (23,258 frames) used for training and 20 sequences (15,069 frames) for testing. The resolution of each frame is 1024×540 pixels. It includes three object categories: car, bus, and truck. On average, each frame contains 18 objects.

TinyPerson: This dataset is specifically designed for tiny pedestrian detection and consists of images collected by UAV-mounted cameras in coastal environments. The objects are extremely small, typically smaller than 20×20 pixels. The dataset contains 1,610 images in total, with 794 used for training and 816 for testing, annotated with 72,651 bounding boxes. Most images have a resolution of 1920×1080 pixels, and each image contains an average of 25 objects.

To provide a clearer understanding of the differences among these datasets, we conduct a quantitative comparison based on four characteristics: average object density, object scale distribution, background complexity (measured by texture entropy), and occlusion rate. The comparison results are summarized in Table 1. These dataset-level statistics offer insights into their structural differences, which are relevant to the evaluation of detection algorithms.

To evaluate the performance of our method, we adopt mAP, AP₅₀, and AP₇₅ as evaluation metrics. Specifically, mAP refers to the mean Average Precision computed across IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, while AP₅₀ and AP₇₅ represent the average precision at IoU thresholds of 0.5 and 0.75, respectively. In addition, to assess detection performance across objects of different scales, we employ metrics such as AP_s, AP_m, and AP_l. Specifically for the TinyPerson dataset, we adopt three scale-specific indicators—tiny1, tiny2, and tiny3—which correspond to pedestrian objects with bounding box sizes in the pixel ranges of (2, 8), (8, 12), and (12, 20), respectively. To further verify the efficiency of our approach, we report the average FLOPs (G) per image on the validation set and the inference speed in frames per second (FPS) as indicators of computational complexity and real-time performance, respectively.

Table 1

Quantitative comparison of dataset characteristics among UAVDT, VisDrone, and TinyPerson.

Dataset	Object Density (avg objects / image)	Scale Distribution (%) (small : medium : large)	Background Complexity (texture entropy)	Occlusion Rate (%) (no : partial : heavy)
VisDrone	53	60.72 : 33.88 : 5.4	3.17	48.75 : 41.45 : 9.81
UAVDT	18	61.26 : 37.08 : 1.66	3.03	88.17 : 8.49 : 3.34
TinyPerson	25	85.67 : 13.38 : 0.94	2.89	–

4.2. Implementation details

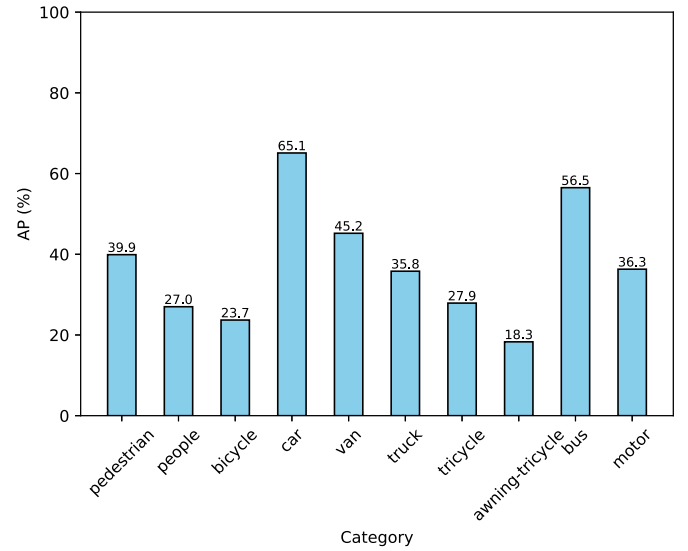
Our work is implemented using the publicly available MMDetection. All input images are resized to a uniform resolution of 1024×640 , and data augmentation techniques are applied to increase data diversity. These include random horizontal flipping with a probability of 50%, photometric distortions (random adjustments of brightness, contrast, and saturation), and random cropping. During training, we use SGD as the optimizer with an initial learning rate of 0.000625, momentum of 0.9, and weight decay of 0.0001. The batch size is set to 2, and the model is trained for 240 epochs. A linear warm-up strategy is adopted for the first 1000 steps, and the learning rate is decayed by a factor of 0.1 at the 90th and 120th epochs.

All experiments are conducted on an NVIDIA RTX 4090 GPU. During inference, our method consumes approximately 3714 MB of GPU memory (VRAM), which remains modest considering the use of high-resolution inputs and a two-stage architecture. Given that modern embedded platforms for UAV deployment (e.g., NVIDIA Jetson Xavier NX, Orin Nano) are typically equipped with 8GB or more of memory, the VRAM usage of our model is within a practical range for real-world deployment.

4.3. Experimental results

To comprehensively evaluate the performance of our proposed method, we compare it with several mainstream detection approaches on the VisDrone dataset. Table 2 presents the detailed experimental results, where the best scores are bolded. As shown in the upper part, several advanced general-purpose object detectors are listed, including Cascade R-CNN (Cai & Vasconcelos, 2018) and CenterNet. As shown, our method achieves improvements of 14.2%, 9.6% in mAP over these methods, respectively. Significant gains are also observed in other metrics, indicating that our proposed coarse-to-fine detection framework is more effective and efficient. To ensure a fair comparison, we also compare our approach with detectors that follow the “filter-then-focus” detection paradigm, as shown in the lower part. The results demonstrate that our model consistently outperforms other methods, achieving the best performance with 37.4% mAP and 30.2% AP_s . Although focus-based re-detection methods typically incur higher computational costs, our framework mitigates this issue by reusing the detector itself to adaptively search for dense object regions. This strategy significantly reduces the required FLOPs, resulting in higher efficiency. It is also worth noting that our model achieves substantial improvements in small and medium objects detection (30.2%, 47.3%), while improvements for large objects are less pronounced. This is primarily because our method is based on image cropping. Although we enlarge each crop by a factor of 1.2 to prevent truncation of medium and large objects, a slight drop in precision is still observed. Nevertheless, this trade-off is acceptable, as small objects dominate UAV imagery, and enhancing their detection performance greatly contributes to the overall accuracy of the model.

In terms of inference speed, we also report the FPS for all compared methods. As shown in Table 2, many existing “filter-then-focus” detectors suffer from low inference speed due to additional components or density estimation networks. For example, AMRNet and CDMNet run at only 2.0 FPS and 2.9 FPS, respectively. In contrast, our method achieves a competitive speed of 3.1 FPS while delivering the best detection performance among all models. Additionally, we report a lightweight vari-

**Fig. 5.** Class-wise AP scores on the VisDrone validation set.

ant of our model, DG-TSOD[†], which performs only single-stage inference without density-guided refinement. Despite its simplicity, this variant achieves 34.8% mAP and 26.4% AP_s , outperforming most general-purpose detectors and even surpassing several filter-then-focus methods, such as CDMNet and CZ Det. Notably, it runs at 14.7 FPS, demonstrating that our framework maintains strong performance even under strict runtime constraints.

We analyze the per-category detection performance on the VisDrone validation set to better understand how our method performs across different object types. As illustrated in Fig. 5, the detection accuracy varies significantly across categories. Our model achieves the highest AP for the “car” (65.1%) and “bus” (56.5%) categories, which typically exhibit larger scales and clearer structures. In contrast, lower AP values are observed for small or heavily occluded categories such as “awning-tricycle” (18.3%), “bicycle” (23.7%), and “people” (27.0%). Despite these variations, the AP for “pedestrian” reaches 39.9%, indicating that our method maintains good performance on challenging small-object categories.

Table 3 presents the quantitative comparison results on the UAVDT dataset. Although our model does not achieve the best performance on this dataset, it still demonstrates highly competitive detection results. In particular, our method achieves 36.0% AP_{50} and 16.0% AP_s , indicating good accuracy and stability despite a performance gap compared to the best-performing model, ESOD. To further understand this performance gap, we conduct a quantitative analysis of the three datasets in terms of object density, scale distribution, background complexity, and occlusion rate (see Table 1). The results show that UAVDT has the lowest average object density (18 objects per image). In addition, 88% of the targets in UAVDT are unoccluded, and the background texture complexity is relatively low. More importantly, UAVDT was originally designed for object tracking and is constructed by extracting image frames from 30 FPS video sequences. This results in high inter-frame redundancy and limited visual diversity and structural complexity. These factors limit the effectiveness of our density-guided refinement module, which is specifi-

Table 2

Performance comparison with mainstream methods on the VisDrone dataset. DG-TSOD (Ours)[†] indicates the results of DG-TSOD using single-stage inference only, without applying density-guided refinement.

Method	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	FLOPs(G)	FPS
Faster RCNN (Ren et al., 2015)	21.8	41.8	20.1	11.9	34.8	55.5	285	18.2
Cascade RCNN (Cai & Vasconcelos, 2018)	23.2	39.9	24.7	15.3	35.6	38.4	362.8	–
CenterNet (Duan et al., 2019)	27.8	47.9	27.6	21.3	42.1	49.8	–	5.6
DetectoRS* (Qiao et al., 2021)	26.8	43.2	28.0	17.5	38.2	41.7	–	3.9
SDP (Ma et al., 2023)	30.2	52.5	30.6	22.6	39.6	39.8	–	21.8
CEASC (Du et al., 2023)	28.7	50.7	28.4	–	–	–	150.2	26.9
DG-TSOD(Ours) [†]	34.8	58.8	35.4	26.4	46.3	53.2	153	14.7
ClusDet (Yang et al., 2019)	28.4	53.2	26.4	19.1	40.8	54.4	436.0	1.8
DMNet (Li et al., 2020)	28.5	48.1	29.4	20.0	39.7	57.1	471.4	2.1
CDMNet (Duan et al., 2021)	29.2	49.5	29.8	20.8	40.7	41.6	421.6	–
GLSAN (Deng et al., 2020)	30.7	55.4	30.0	–	–	–	–	–
QueryDet (Yang et al., 2022)	28.3	48.1	28.8	–	–	–	888.4	2.7
AMRNet (Wei et al., 2020)	31.7	52.7	33.1	23.0	43.4	58.1	–	–
CZ Det (Meethal et al., 2023)	33.2	58.3	33.1	26.1	42.6	43.4	456.4	8.4
DNTR (Liu et al., 2024b)	34.4	57.9	35.3	–	–	–	–	–
ESOD (Liu et al., 2024c)	36.0	59.7	–	28.3	–	–	119.5	36.4
DG-TSOD(Ours)	37.4	61.7	38.6	30.2	47.3	52.0	305.9	3.1

Table 3

Performance comparison with mainstream methods on the UAVDT dataset.

Method	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l	FLOPs(G)
Faster RCNN (Ren et al., 2015)	12.1	23.5	10.8	8.4	21.5	14.7	249.9
ClusDet (Yang et al., 2019)	13.7	26.5	12.5	9.1	25.1	31.2	421.7
DMNet (Li et al., 2020)	14.7	24.6	16.3	9.3	26.2	35.2	575.3
CDMNet (Duan et al., 2021)	16.8	29.1	18.5	11.9	29.6	15.7	–
GLSAN (Deng et al., 2020)	17.1	28.3	18.8	–	–	–	–
AMRNet (Wei et al., 2020)	18.2	30.4	19.0	13.3	33.5	–	–
CEASC (Du et al., 2023)	17.1	30.9	17.8	–	–	–	64.1
ESOD (Liu et al., 2024c)	22.5	40.7	–	–	–	–	41.1
RemDet-L (Li et al., 2025)	20.6	34.5	22.1	13.9	31.4	30.3	–
DG-TSOD(Ours)	21.4	36.0	23.7	16.0	33.2	35.4	284.6

Table 4

Performance comparison with mainstream methods on the TinyPerson dataset.

Method	AP ₅₀ ^{tiny}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{tiny2}	AP ₅₀ ^{tiny3}	AP ₅₀ ^{small}
RetinaNet (Lin et al., 2017b)	33.5	12.2	38.8	47.4	48.3
Faster RCNN (Ren et al., 2015)	47.3	30.2	51.6	58.9	63.2
FCOS (Tian et al., 2019)	17.9	2.88	12.9	31.1	40.5
SM (Yu et al., 2020)	51.3	33.9	55.2	62.6	67.0
SM+ (Jiang et al., 2021)	52.6	34.2	57.1	63.3	67.4
Cascade RCNN (Cai & Vasconcelos, 2018)	57.2	45.2	60.1	65.1	70.7
ESOD (Liu et al., 2024c)	61.3	–	–	–	74.4
DG-TSOD(Ours)	61.5	49.6	62.6	70.2	74.8

cally designed to improve small object detection in crowded scenes with occlusions and complex backgrounds.

In addition, we compare the performance of different mainstream methods on the TinyPerson dataset. As shown in Table 4, our model achieves superior performance across all evaluation metrics, particularly in detecting extremely small objects (i.e., objects with bounding box sizes in the range of [2, 20] pixels), reaching 61.5 %, which significantly surpasses other methods. It is worth noting that although some images in the TinyPerson dataset are not captured by UAVs, the dataset contains a high proportion of small objects. This validates the strong generalization capability of our proposed method across various scenarios, as well as its robustness and effectiveness in small object detection tasks.

4.4. Ablation studies

To evaluate the contribution of each module to small object detection performance, we design and conduct a series of ablation experiments. All experiments are performed on the VisDrone dataset to validate the effectiveness of our method and the performance gains introduced by each component.

Effect of DOR Seeker. To address the problem of uneven object distribution in UAV images, we propose DOR Seeker. This module automatically identifies and localizes dense object regions within an image, allowing the detector to focus on high-density areas. As illustrated in Fig. 6, we visualize the entire detection pipeline of our method. In subfigure (a), the binarized heatmap highlights the spatial distribution of dense object regions, which are identified by the DOR Seeker. This visualization clearly demonstrates that DOR Seeker can effectively localize areas with high object concentration. Subfigure (b) shows the initial coarse detection results. Without refined detection on the dense regions, the detector fails to capture fine details of the objects, especially missing several small objects. In contrast, Subfigure (c) shows the refined detection results after further processing of the dense regions, where previously missed objects are successfully detected and localized with higher precision.

Hyperparameters in DOR Seeker. We conduct an ablation study on the hyperparameter τ used in the DOR Seeker, as shown in Fig. 7. The detection performance reaches its peak when the threshold τ is set to 1, achieving a mAP of 37.4 %. However, as τ increases, the detection performance gradually degrades. We attribute this decline to the fact

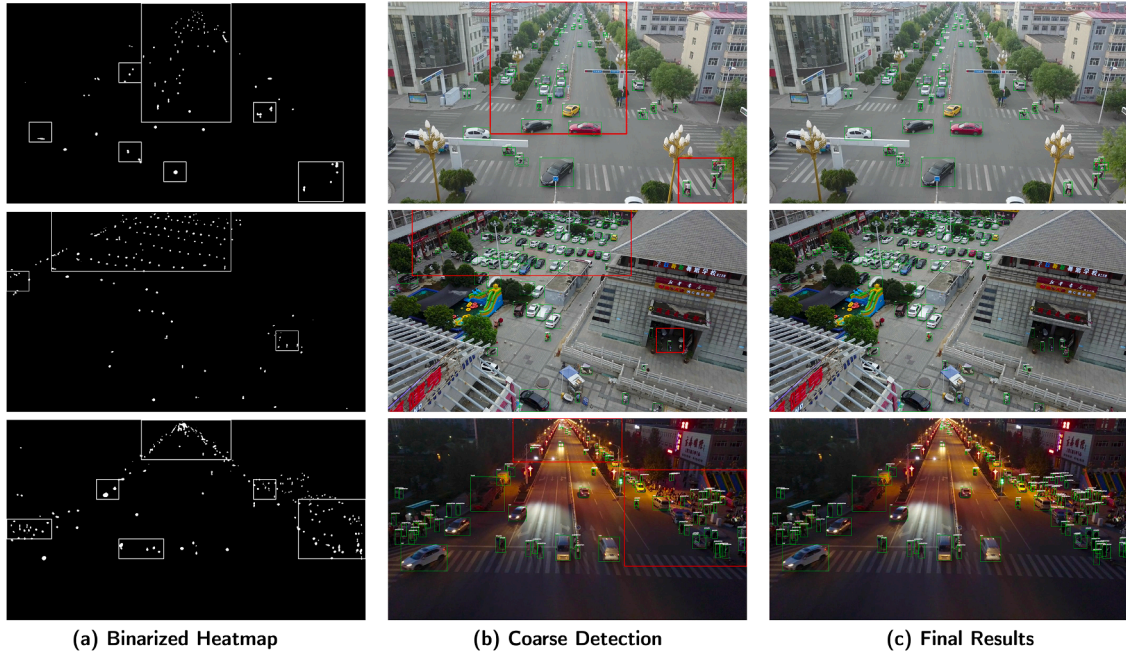


Fig. 6. Visualization of density-based cropping and detection. (a) the binarized heatmap highlighting high-density regions. (b) the coarse detection results obtained in the first stage. (c) the final detection results after second-stage processing.

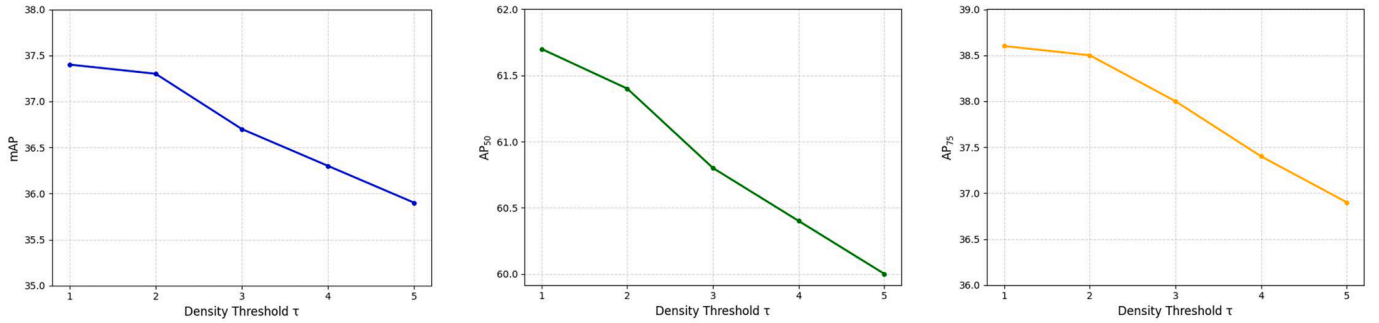


Fig. 7. Performance comparison under different density thresholds.

that a higher threshold may exclude some low-density object regions, leading to missed detections and reduced overall accuracy. This observation suggests that the choice of the density threshold in the cropping strategy must strike a balance between improving the detection accuracy for small objects and avoiding excessive cropping. Proper tuning of this parameter is crucial for achieving optimal detection performance.

We further investigate the impact of two additional key hyperparameters in the DOR Seeker: the number of crops K and the grid size used for density estimation. As shown in Table 5, increasing the number of selected crops K from 0 (no refinement) to 2 leads to a significant improvement in detection performance, with mAP increasing from 34.8% to 37.4% and AP_s from 26.4% to 30.2%. This confirms that introducing a small number of refined crops allows the model to focus on dense object regions, thereby improving small object recall. However, when K is further increased to 3, the performance saturates, and FPS drops from 3.1 to 2.7. This suggests that using too many crops not only incurs additional computational cost but also introduces redundant regions that provide limited new information. Thus, setting $K = 2$ achieves the best trade-off between performance and inference speed.

In addition, we analyze the effect of different grid size settings in Table 6. A coarser grid (8, 5) results in insufficient spatial granularity, causing many local dense regions to be merged or missed, leading to suboptimal detection (mAP = 36.9%). On the other hand, an excessively fine grid (32, 20) increases computational overhead and may gen-

Table 5

Performance under different values of the crop number K .

K	mAP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	FPS
0	34.8	58.8	35.4	26.4	46.3	53.2	14.7
1	37.2	61.5	38.4	29.9	47.2	52.1	3.3
2	37.4	61.7	38.6	30.2	47.3	52.0	3.1
3	37.4	61.6	38.6	30.2	47.3	51.9	2.7

Table 6

Performance under different grid size settings.

Grid size	mAP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	FPS
(8,5)	36.9	61.2	37.9	29.6	47.0	51.5	3.2
(16,10)	37.4	61.7	38.6	30.2	47.3	52.0	3.1
(32,20)	36.2	60.3	37.4	28.8	46.6	52.9	2.3

erate fragmented density maps, negatively affecting both mAP (36.2%) and FPS (2.3). The best performance is obtained at a moderate grid setting of (16, 10), achieving the highest mAP of 37.4% and AP_s of 30.2%, while maintaining a reasonable inference speed of 3.1 FPS.

Effect of Density-Based Cropping. We apply density-based cropping during both the training and inference phases to achieve optimal performance. Specifically, during training, the density-cropped regions

Table 7

Detection results when density-based cropping is applied during both training and inference.

Train	Test	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
		33.9	57.1	34.4	25.6	43.6	51.3
	✓	35.7	59.6	36.9	28.4	45.6	50.7
✓		34.8	58.8	35.4	26.4	46.3	53.2
✓	✓	37.4	61.7	38.6	30.2	47.3	52.0

Table 8

Comparison of detection performance with different head module designs.

Method	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
–	34.6	58.5	35.7	28.2	45.4	47.3
EMSCnv	36.5	61.1	37.4	29.9	46.3	49.0
Dcnv	36.2	60.4	37.5	29.8	46.2	48.7
EMSCnv + Dcnv	37.4	61.7	38.6	30.2	47.3	52.0

are first rescaled and then incorporated into the data augmentation pipeline together with the original training images. During inference, a two-stage strategy is adopted: the first stage performs detection on the entire image, and the second stage refines the results by focusing on the crops. As shown in Table 7, using density-based cropping only during inference yields an mAP of 35.7%. When applied during both training and inference, the mAP increases to 37.4%, reflecting a 1.7% absolute improvement. In addition, consistent performance gains are observed in AP₅₀, AP₇₅, and across different object scales (AP_s, AP_m, and AP_l). These results demonstrate that adopting density-based cropping in both phases significantly boosts overall performance, particularly in detecting small objects.

Effect of the Improved Decoupled Detection Head. To improve localization accuracy, we introduce the EMSCnv and Dcnv modules into the regression branch of the decoupled detection head. The EMSCnv module, through multi-scale convolutional operations, effectively captures fine-grained details of small objects. In scenarios involving occlusion, larger convolution kernels can extract contextual information outside the occluded regions, thereby mitigating the negative impact of occlusion on detection accuracy. When dense object regions are incomplete or affected by occlusions, the Dcnv module dynamically adjusts the sampling locations of the convolutional kernels. This enables the network to shift sampling points from occluded areas to more informative regions, allowing for greater flexibility in adapting to variations in object shape and position, ultimately improving detection performance. In Table 8, the “–” denotes the baseline decoupled detection head using only standard convolutional blocks without the proposed enhancements. Experimental results show that integrating EMSCnv and Dcnv leads to significant improvements in detection accuracy, particularly for small and large objects, demonstrating the effectiveness of the proposed enhancements.

Cross-scale Detection Trade-off Analysis. To further evaluate the impact of our cropping-based strategy on objects of different scales, we conduct a detailed analysis of detection performance across small, medium, and large objects. As shown in Table 9, our method achieves the best overall performance when the crop expansion ratio is set to 1.2, reaching the highest mAP (37.4%) and AP_s (30.2%). Compared to the baseline setting (1.0), both small and medium object performance are significantly improved, with AP_s increasing from 29.2% to 30.2% and AP_m from 46.6% to 47.3%. This confirms that the proposed density-aware cropping strategy, with a moderate expansion factor, effectively enhances the detector’s ability to localize and recognize small and moderately sized objects. Notably, the detection performance on large objects (AP_l) experiences a slight drop (from 52.6% to 52.0%) as the expansion ratio increases. However, this trade-off is acceptable since large objects account for only 5.4% of all annotated instances in the VisDrone dataset, and the overall improvement in small

Table 9

Performance under different crop expansion ratios.

Expansion Ratio	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
1.0	36.5	60.5	37.7	29.2	46.6	52.6
1.2	37.4	61.7	38.6	30.2	47.3	52.0
1.4	37.2	61.1	38.4	30.2	47.0	52.3
1.5	36.9	60.8	38.1	30.0	46.7	53.7
2.0	35.7	58.6	37.3	29.0	45.4	51.3

Table 10

Performance comparison of different loss functions.

Method	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
$\mathcal{L}_1$	36.2	61.4	37.3	28.8	46.5	54.2
NWD	37.1	61.5	38.2	29.6	47.1	51.4
NWD + $\mathcal{L}_1$	37.4	61.7	38.6	30.2	47.3	52.0

and medium object accuracy contributes more significantly to the final mAP.

To gain deeper insights into cross-scale performance behavior, we further present normalized confusion matrices for small, medium, and large targets in Fig. 8. It can be observed that small targets have a higher rate of misclassification into neighboring classes, particularly among visually similar vehicle types such as “car” and “van”, “awning-tricycle” and “tricycle”. Our refinement strategy helps mitigate these confusions, especially in the small and medium categories. In contrast, large targets are generally recognized with higher confidence and less confusion, although a few classes like “awning-tricycle” and “motor” still exhibit notable ambiguity due to their low frequency and occlusion variability. These results demonstrate that our method achieves a favorable balance in cross-scale detection: significantly improving small and medium object detection, while maintaining reasonable performance on large targets. This balance is crucial for UAV-based applications, where small and medium-sized objects dominate most scenes.

Different Loss Functions. We improve the regression branch of the detection head by addressing the limitations of the traditional $\mathcal{L}_1$ loss, which is commonly used in object detection. Specifically, to mitigate the sensitivity of $\mathcal{L}_1$ loss to positional deviations in small object detection, we introduce the Normalized Wasserstein Distance (NWD) loss. As shown in Table 10, the use of NWD loss leads to a notable improvement in small object detection performance, with AP_s increasing from 28.8% to 29.6%, demonstrating enhanced robustness in localizing small objects. However, we also observe a 2.8% decrease in the detection accuracy for large objects when using NWD alone, a phenomenon that we have analyzed in detail in Section 3.4. To address this issue and achieve better detection performance across object scales, we propose a weighted loss function that combines NWD loss and $\mathcal{L}_1$ loss. Experimental results show that the weighted loss further improves small object performance, increasing AP_s from 29.6% to 30.2%. In addition, it also yields gains for medium and large objects, with AP_m rising from 47.1% to 47.3% and AP_l from 51.4% to 52.0%. Overall, mAP increases from 37.1% to 37.4%, indicating that the proposed loss function achieves a better balance across objects of different scales.

Effect of Loss Weight Hyperparameters. We investigate the influence of the loss weight hyperparameters λ_1 and λ_2 on the model performance, as shown in Table 11. Overall, the results demonstrate that appropriate balancing between different loss terms can lead to noticeable performance improvements across all evaluation metrics. Among the evaluated configurations, the setting $\lambda_1 = 2$, $\lambda_2 = 2$ achieves the highest mAP (38.2%) and the best performance in all AP metrics, indicating that emphasizing both loss components equally can yield the most favorable results. However, we ultimately adopt $\lambda_1 = 2$, $\lambda_2 = 0.5$ in our main experiments. This decision is based on the principle of emphasizing the effectiveness of our proposed method itself, rather than pursuing the

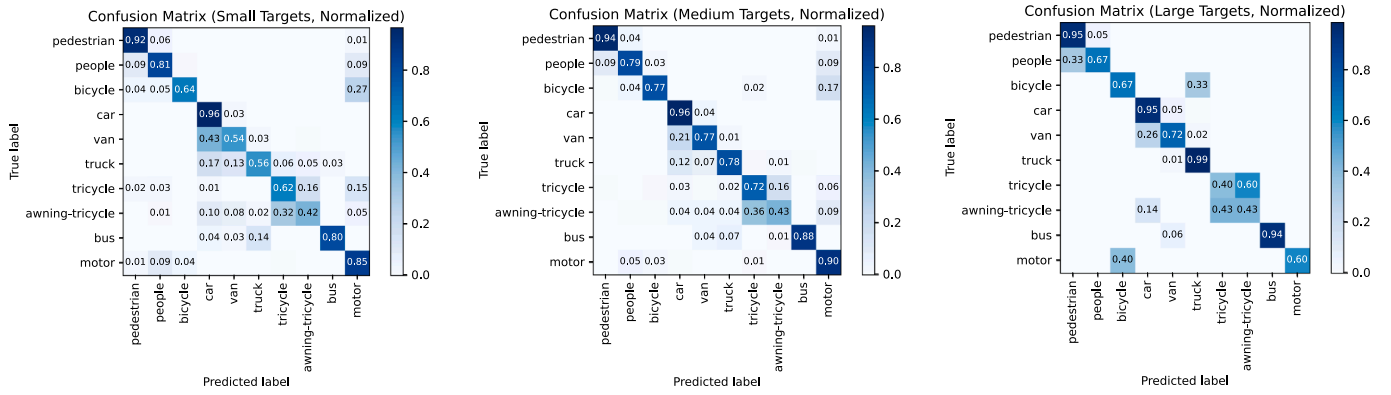


Fig. 8. Normalized confusion matrices for small, medium, and large targets, respectively. Each matrix shows the per-class detection performance under different object sizes. The values are row-normalized to reflect the distribution of predicted labels for each ground-truth class.

Table 11
Performance of the model under different settings of the loss weight hyperparameters.

λ_1	λ_2	mAP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
1	0.5	37.1	61.5	38.3	29.9	47.2	52.0
2	0.5	37.4	61.7	38.6	30.2	47.3	52.0
2	1	37.3	61.3	38.5	30.0	47.1	52.4
1	2	36.5	59.7	38.3	30.0	46.6	52.7
2	2	38.2	62.5	39.9	31.5	47.8	53.5

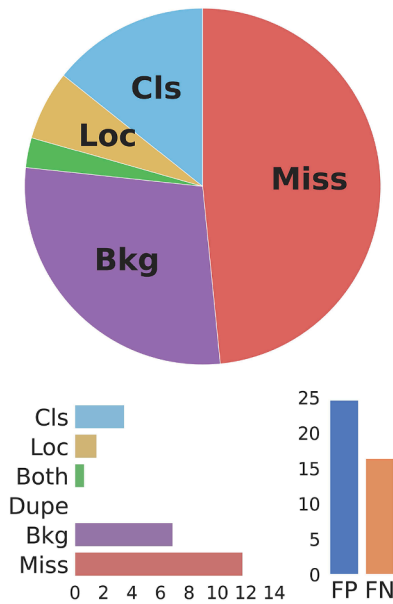


Fig. 9. Error analysis. Cls: localized correctly but classified incorrectly; Loc: classified correctly but localized incorrectly; Both: both classification and localization errors; Dupe: duplicate detection error; Bkg: background falsely detected as foreground; Miss: missed ground truth object.

best result through extensive hyperparameter tuning. This setting still achieves competitive performance (37.4% mAP and 30.2% AP_s), validating the robustness of our framework under reasonable hyperparameter values. It demonstrates that our method can deliver strong results even without exhaustive manual adjustment, highlighting its generality and practicality.

TIDE-Based Error Distribution Analysis. To better understand the performance bottlenecks of the proposed model, we conduct a detailed error analysis on the VisDrone validation set using the TIDE (Bolya et al., 2020) tool. Fig. 9 shows the results; missed detections (Miss) are the

predominant source of false negatives (FN), primarily occurring when the model fails to detect small objects in densely populated regions. Moreover, background confusion (Bkg) and classification errors (Cls) are major contributors to false positives (FP), often arising in cluttered urban scenes where background elements are mistakenly recognized as targets. While localization errors (Loc) and duplicate detections (Dupe) are also present, their overall impact is relatively limited. These results highlight the importance of improving the model's robustness in dense environments and its ability to distinguish true objects from background clutter, which are critical for enhancing UAV-based object detection performance.

5. Conclusion

In this paper, we propose DG-TSOD, a novel UAV object detector specifically designed for small and densely distributed objects in aerial imagery. The framework adopts a coarse-to-fine detection strategy, where a designed DOR Seeker focuses on dense object regions to guide fine-grained detection. An enhanced detection head is integrated to effectively capture contextual information and improve detection accuracy. Furthermore, we introduce a weighted regression loss based on Normalized Wasserstein Distance (NWD), which alleviates the sensitivity of traditional IoU-based losses to positional deviations in small objects, thereby improving the quality of bounding box regression. Experimental results demonstrate that DG-TSOD achieves a favorable trade-off between accuracy and speed. In future work, we plan to perform adaptive upsampling of dense object regions at the feature level, instead of directly feeding upsampled images into the backbone, to further enhance detection performance.

CRedit authorship contribution statement

Bojun Xie: Writing – original draft, Visualization, Validation, Methodology, Conceptualization; **Yanjie Wang:** Writing – original draft, Visualization, Validation, Software, Formal analysis; **Meihui Han:** Validation, Software, Investigation; **Yidan Wang:** Writing – review & editing, Validation, Investigation, Project administration; **Junfen Chen:** Writing – review & editing, Investigation, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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